

Last week : • Erdős-Stone-Simonovits

• Stability method

$$C_4 = K_{2,2}$$

$$\rightarrow \bullet \text{ex}(n, C_4) = \left(\frac{1}{2} + o(1)\right) n^{3/2}$$

$$\bullet \text{KST: } \text{ex}(n, K_{s,t}) = O(n^{2 - 1/s}) \\ 2 \leq s \leq t$$

1. Even cycles from BFS (Breadth First Search)

• Bondy-Simonovits

$$\text{ex}(n, C_{2k}) = O(n^{1 + \frac{1}{k}}) \Leftarrow \begin{matrix} C_4 \\ C_6 \\ C_{10} \end{matrix}$$

Matching lower bound $\Omega(n^{1 + \frac{1}{k}})$

known only for $k = 2, 3, 5$.

Open problem

Improve $\Omega(n^{6/5}) \leq \text{ex}(n, C_8) \leq O(n^{5/4})$

Asymmetric setting

• $\text{ex}(m, n, H) = \max \# \text{edges in a } G \subseteq K_{m,n}$

which is H -free



Known $ex(m, n, C_{2k}) = \begin{cases} O(m^{\frac{1}{k}} (mn)^{\frac{1}{2}} + m + n) & \text{even } k \\ O((mn)^{\frac{1}{2}(1+\frac{1}{k})} + m + n) & \text{odd } k \end{cases}$

Erdős asked whether $ex(n^{2/3}, n, C_6) = O(n)$.
 Disproved. $\geq \Omega(n^{16/15})$

Open problem Is $ex(n^{2/3}, n, C_6) = \Omega(n^{10/9})$?

Thm 1.3 For large n ,

$$ex(n, C_{2k}) \leq 4k \cdot n^{1+\frac{1}{k}}$$

• Recently: improved to $O(\sqrt{k \log k} \cdot n^{1+\frac{1}{k}})$

$$\Omega(n^{1+\frac{1}{k}}) \quad \text{vs} \quad o(n^{1+\frac{1}{k}})$$

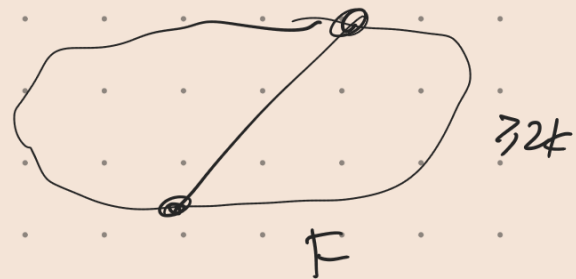
Open problem Is $ex(n, C_{2k}) = o(\sqrt{k} \cdot n^{1+\frac{1}{k}})$?

Upp bd on C_{2k}

Ex 1.5 $\forall G, \exists H \subseteq G \begin{cases} \text{bipartite} \\ e(H) \geq e(G)/2 \end{cases}$

Ex 1.6 $\forall G, \exists H \subseteq G \text{ s.t. } \delta(H) \geq d(G)/2$

Def: Θ_k -graph is a cycle of length $\geq 2k$ with a chord.



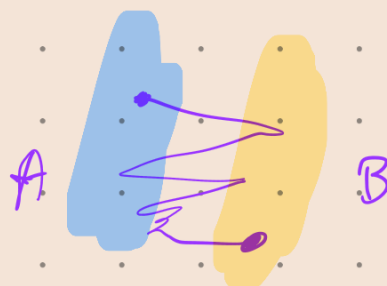
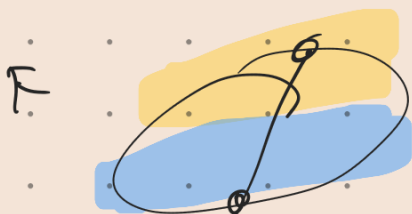
Ex 1.8 $\forall k \geq 3, \forall \text{ bip } H$

$d(H) \geq 2k \implies \exists \Theta_k\text{-graph} \subseteq H$

Θ_k -graph has paths of many distinct lengths.

Lem 1.9 $F: \Theta_k\text{-graph}$, take non-trivial $A \cup B = V(F)$

zf F is not bipartite with bipartition $A \cup B \implies \exists A, B\text{-path of all lengths less than } |F|$.

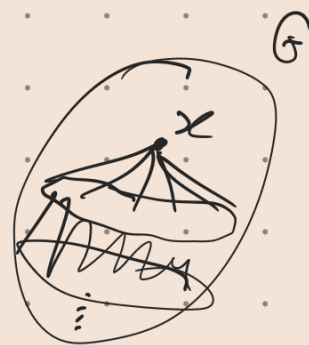


Idea for bounding $ex(n, G_k) = O(n^{1+1/k})$



BFS

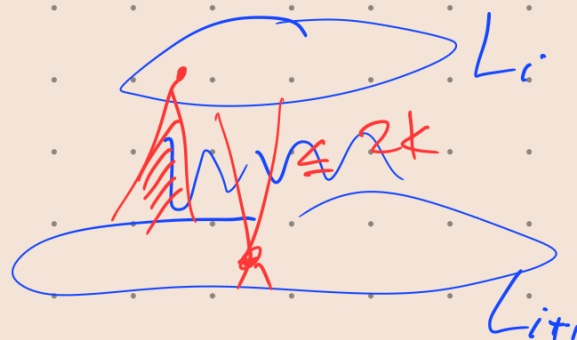
Breadth
First
Search



Lem 1.9 $\Rightarrow$ $[L_i, L_{i+1}]$ is Θ_k -free $\forall i \leq k-1$

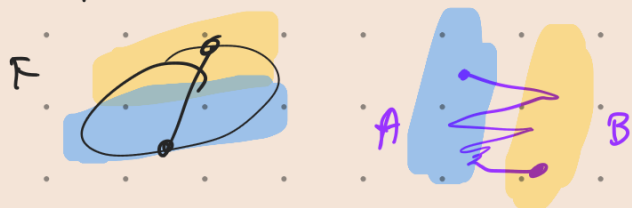
Ex 1.8 $\Rightarrow$ sparse

$$\Rightarrow \frac{|L_{i+1}|}{|L_i|} = O(d/k)$$



Lem 1.9 $F: \Theta_k$ -graph, take non-trivial $A \cup B = V(F)$

z.f. F is not bipartite with bipartition $A \cup B \Rightarrow \exists A, B$ -paths of ^{all} lengths less than $|F|$.

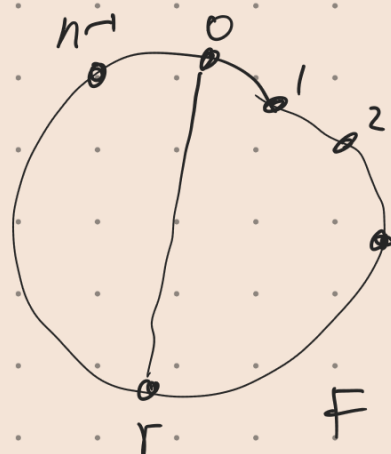


pf. (Sketch) Fix $l < n = |F|$. Suppose there is

no A, B -path of length $l \Rightarrow F$ is bip. & $A \cup B$ is a bipartition

- $V(F) = \mathbb{Z}_n$

- think of $A \cup B$ as $\mathbb{Z}$ -colouring C of $\mathbb{Z}_n$.



Def: $P = \{m \in \mathbb{Z}_n : \forall i \in \mathbb{Z}_n \quad C(i) = C(i+m)\}$

So $l \in P$.

- $A \cup B$ is non-trivial $\Rightarrow$ the smallest period $m \in P$ satisfies $m \nmid n$.

$\leadsto P = \{mi : i \in \mathbb{Z}_n\}$

- May assume $m > 2$, (o.w. $A \cup B$ is bipartition)

$\Rightarrow$ if $n-r \equiv r \equiv 1 \pmod m$

$\Rightarrow n \equiv 2 \pmod m \Rightarrow m \nmid n$

Say $r-1 \equiv 0 \pmod m$

$r-1 \notin P$, $l \in P$



$$j+l+r-1 \Rightarrow \exists j \in \mathbb{Z}_n \text{ s.t. } \underline{c(j)} \neq \underline{c(j+r-1+l)}$$

$$\cdot \quad \underline{j, j+r, \dots, -1, 0, r, 2r, \dots, j+l+r-1}$$

pf $ex(n, G_k) \leq 4k \cdot n^{1+\frac{1}{k}}$

$$\cdot \text{ Let } |G|=n, \quad e(G) \geq 4k \cdot n^{1+\frac{1}{k}} \Rightarrow G_k \subseteq G$$

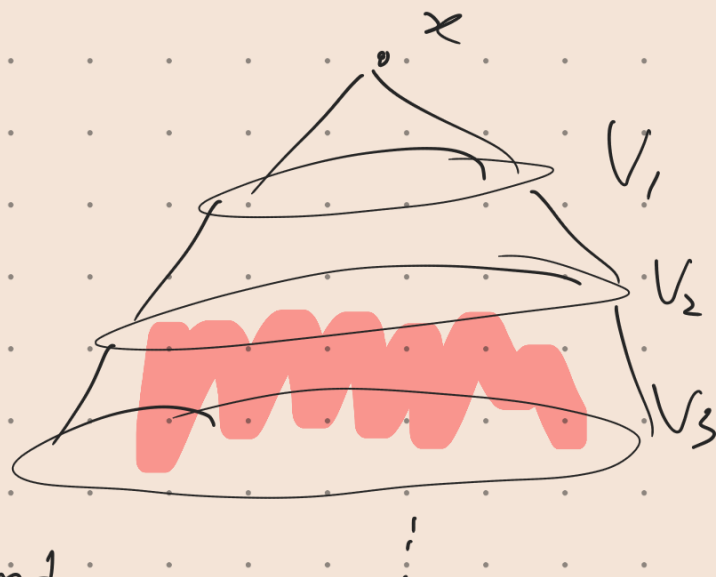
By ex 1.5 & 1.6 bipartite
pass to subgraph H with $\delta(H) \geq \frac{d(G)}{4} \geq 2k \cdot n^{\frac{1}{k}}$

$$\cdot \text{ Take } x \in V(H), \quad V_i = N_H^i(x)$$

$$H_i = H[V_i, V_{i+1}]$$

Claim $\forall i \leq k-1$

H_i has no G_k -subgraph.



pf (Claim) Suppose H_i has a G_k -subg F

with bipartition Y & Z

- Let y = minimal ancestor of Y_i

- z = child of y s.t. $\text{dist} = i-l$

$$\begin{cases} D(z) \cap Y_i \neq \emptyset \\ Y_i \setminus D(z) = \emptyset \end{cases}$$

$$Y_i \setminus D(z) = \emptyset$$

- Let $B = (Y \cup Z) \setminus A$

- Let $l = \text{dist}(x, y) < i$

$$\Rightarrow 2k - 2(i-l) < 2k \leq |F|$$

- Lemma 1.9 $\Rightarrow \exists$ a, b-path P of length

$$\frac{2k - 2(i-l)}{\text{even}}, \quad a \in A, \quad b \in B$$

$\Rightarrow$

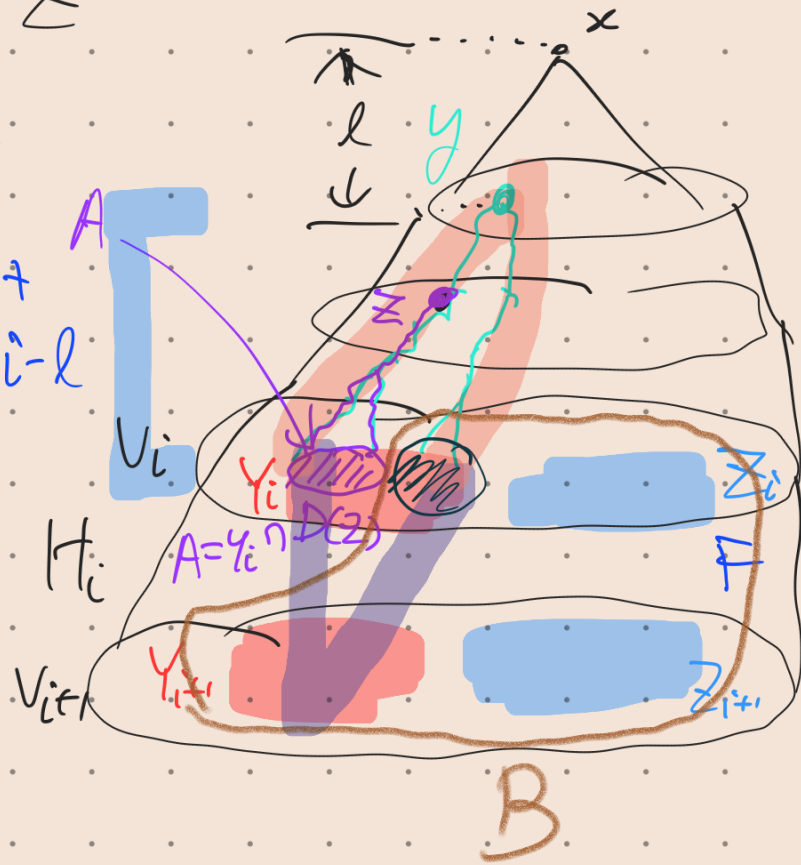
$$b \in Y$$

$$b \in V_i$$

$\Rightarrow$

$$b \in Y_i \setminus A$$

P with a, y -, b, y -paths $\Rightarrow C_{2k} \subseteq \mathbb{R}$



Supersaturation

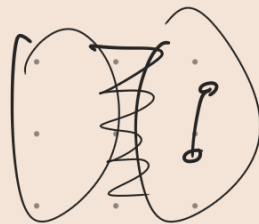
By defn : n -vx G with $ex(n, H) + 1$

edges $\rightsquigarrow \exists$ one copy
 H in G

Supsat { H bip $\rightsquigarrow$ Sidorenko conj
nonbip $\rightsquigarrow$ Erdős-Rademacher

Prop (Rademacher)

G { n -vx
 $\frac{n^2}{4} + 1$ edges $\Rightarrow \exists \lfloor \frac{n}{2} \rfloor$ K_2 in G

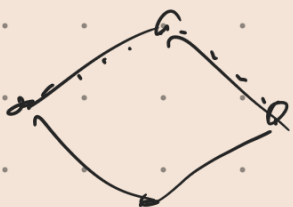


• Will see $\forall G$ w/ $2ex(n, C_4)$
 $\rightsquigarrow$ lots of C_4

Thm 2.2 n -vx G

$d(G) \geq 2\sqrt{n} \Rightarrow \frac{d(G)^4}{8} C_4$ in G .

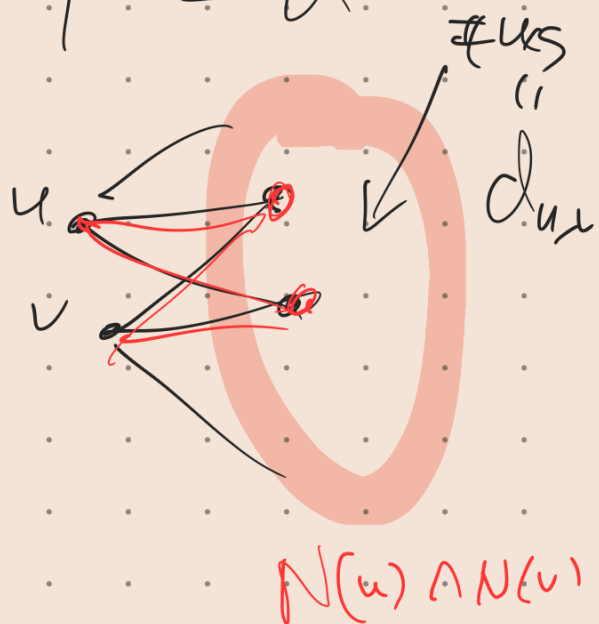
Rmk $G(n, p)$, $p = \frac{d \cdot n}{\binom{n}{2}} \approx \frac{d}{n}$



$$n^4 p^4 \approx d^4$$

pf: Idea:

$$\# C_4 = \sum_{u, v} \binom{d_{u,v}}{2}$$



Jensen

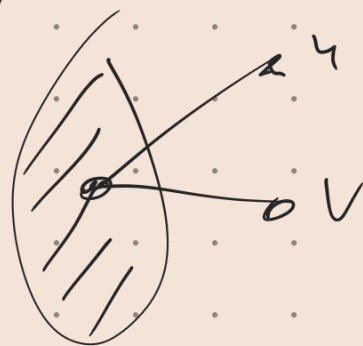
$$\geq \binom{n}{2} \binom{d_{co}}{2}$$

d_{co} = ave codeg

$$= \frac{1}{\binom{n}{2}} \sum_{u, v} d_{u, v}$$

$$d_{co} = \frac{\# K_{1,2}}{\binom{n}{2}}$$

cherries



$$\# K_{1,2} = \sum_v \binom{d(v)}{2}$$

Jensen

$$\geq n \binom{\frac{1}{n} d(v)}{2}$$



Regularisation

Lemma 3.1 n -ux G $e(G) \geq c \cdot n^{1+\epsilon}$

$\implies \exists H \subseteq G, |H| = m \geq n^{\Omega(1)}$

$$\& \quad e(H) \geq \frac{2c}{5} \cdot m^{1+\epsilon}$$

$\left\{ \begin{array}{l} K\text{-almost reg} \end{array} \right., \quad K = 20 \cdot 2^{\frac{1}{\epsilon^2} + 1}$

$$\Delta(H) \leq K \cdot \delta(H)$$

Cube & degeneracy.

Def: G k -degenerate:

$$\text{i.e. } \forall H \subseteq G, \delta(H) \leq k$$



Conj 4.1 H is k -degenerate

$$\Rightarrow ex(n, H) = O(n^{2 - 1/k})$$

Best : $ex(n, H) = O(n^{2 - 1/(4k)})$

Conj 4.2 H be $K_{k,k}$ -free

bipartite with max-deg k

on one side



$K_{k,k}$

$$\Rightarrow ex(n, H) = O(n^{2 - 1/k - \Omega(1)})$$



Open problem

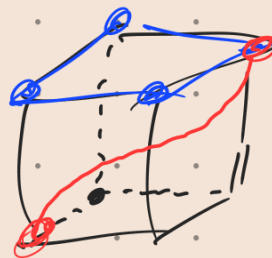
Q_k

k -dim cube

is $K_{3,3}$ -free.

$$ex(n, Q_k) \leq O(n^{2 - 1/k - \Omega(1)})$$

Q_3



True if $k=3$.

$$\Omega(n^{3/2}) \leq ex(n, Q_3) \leq O(n^{8/5})$$

Open: Improve $\nearrow$

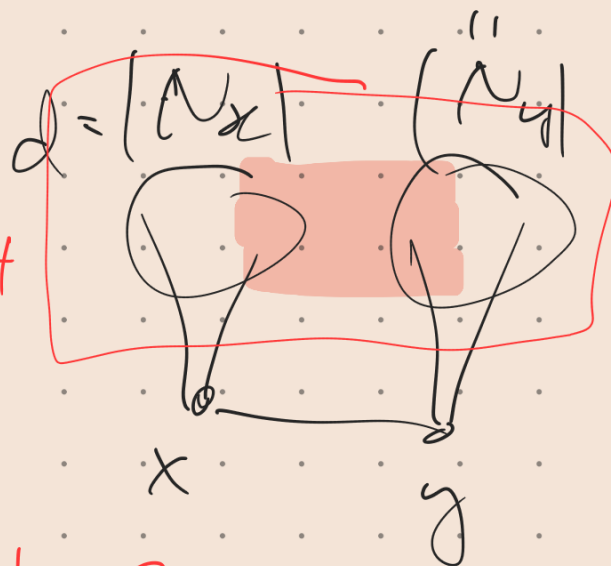
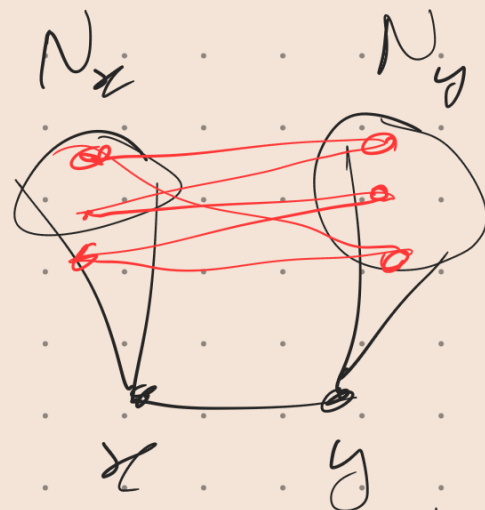
Idea

Q_3^+

Use supsat for C_4

$$e(G) \geq C \cdot n^{8/5}$$

$$\Rightarrow \exists xy \in E(G)$$



$$e(H) > ex(|H|, G)$$

